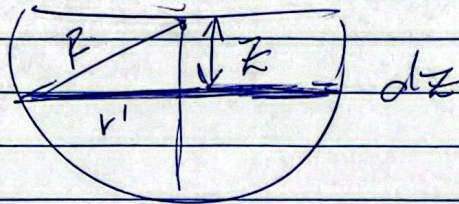


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Problema #5

$$W = \vec{F} \cdot d\vec{l}$$

Consideremos discos de grosor dz y radio r' como se muestra en la fig.



$$R^2 = z^2 + r'^2$$

$$r'^2 = R^2 - z^2$$

$$W = \vec{P} \cdot \vec{z}$$

$$dW = dP \cdot z$$

$$= dm \cdot g \cdot z$$

$$= dV \cdot g \cdot z$$

$$\rho = \frac{dm}{dV} = 1$$

$$\Leftrightarrow dm = dV$$

El volumen de cada disco es

$$A \cdot h = \pi r'^2 \cdot dz$$

$$= \pi (R^2 - z^2) dz$$

Entonces

$$dW = \pi (R^2 - z^2) dz \cdot g \cdot z$$

$$W = g\pi \int_0^R z(R^2 - z^2) dz$$

$$W = g\pi \left(\frac{R^2 z^2}{2} - \frac{z^4}{4} \right) \Big|_0^R =$$

$$\boxed{\frac{g\pi R^4}{4}}$$

(b) potencia

$$P = \frac{W}{t}$$
$$\boxed{P = \frac{g\pi R^4}{4t}}$$